

NHIS Assistant Prototype

Draft management summary for demo discussion

Purpose

The NHIS Assistant prototype demonstrates a conversational, widget-style interface for NHIS users. It answers plain-language adult and child Summary Health Statistics estimate questions from controlled DHIS/NHIS data files, routes general NHIS questions to approved source material, and provides optional GPT-style wording/follow-up support while keeping estimates deterministic and source-grounded.

Current demo capabilities

Capability	What it demonstrates
Estimate Q&A	Adult/child SHS estimates, CIs, special-code footnotes, latest-year logic, multi-year responses, and subgroup summaries.
Source-grounded FAQ	General NHIS, public-use-file, participant, privacy, and “why participate” answers using a controlled local index.
Interactive follow-up	Conversation context supports follow-ups such as “by SVI,” “last 2 years,” “what about kids,” and “where do I get the data?”
UI options	Draft production view, compact iframe/embed view, and separate debug/reviewer view with feedback logging.

Important guardrails

- The model does not calculate estimates or override suppression/reliability rules.
- Unavailable estimates trigger fallback/source guidance rather than invented statistics.
- Teen estimate questions redirect to the NHIS Teen SHS tool; teen participation questions route to resources.
- OpenAI/GPT support is optional and only used for wording, follow-up handling, and structured interpretation.

Example demo questions

- What percent of adults had diabetes last year by SVI?
- What percent of kids got a flu shot last year?
- What percent of gay men had current asthma last year?
- Why should teens participate in NHIS?
- Where can I find the 2024 NHIS public use files?
- How has NHIS data helped with diabetes?

Recommended next steps before production consideration

- Governance/source review: confirm approved estimate files, participant resources, fallback wording, and citation rules.
- Technical hardening: define hosting, logging, feedback storage, API-key handling, and deployment process.
- UI/accessibility review: align with CDC web style, mobile behavior, and Section 508/accessibility requirements.
- Retrieval expansion: build a production-grade approved NHIS documentation/resource index and refresh workflow.
- Testing: run batch regression tests using real reviewer questions before broader release.

Source basis for current demo

The local resource index includes approved NHIS participant links and provided participation/impact materials. These materials describe NHIS as a trusted data source used by outside organizations to improve medical care, health programs, access to devices/medicine, and quality of life, while noting that NCHS/NHIS do not make policy recommendations or engage in advocacy. Example benefit areas include cancer screening, asthma, diabetes education, insulin costs, hearing aids, and anxiety screening.

Prototype status: demo only; not a final production release.